

# Existence of a True Power

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## 1.1 The Recurrence and Fixed-Point Equation

Let us recall the recurrence for  $Y_n$

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left( Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right) + J^{(n)} \max \left( Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), n \geq 1, \quad Y_1 = 1, Y_2 = 0.$$

Because our analysis concerns only the sequence of marginal distributions  $\mathcal{L}(Y_n)$ , we can use the same random variables  $I, J$  for each  $n$ . This reuse does not impact the distributional recurrence, provided that at each step  $n$ , the inputs  $Y_{U_n}^{(1)}$  and  $Y_{n+1-U_n}^{(2)}$  are drawn as conditionally independent copies from the current distribution  $\mathcal{L}(Y_n)$ , independent of  $I, J$ , and  $U_n$ . So the recurrence can be rewritten as

$$Y_{n+2} \stackrel{d}{=} I \left( Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} \right) + J \max \left( Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), n \geq 1, \quad Y_1 = 1, Y_2 = 0. \quad (1.1)$$

For the same reason, we set  $U_n = \lceil nU \rceil$  for every  $n$ , where  $U$  is uniformly distributed on  $[0, 1]$ . Let  $X_n = Y_n/n^\alpha$ , for some  $\alpha \in (0, 1]$ . We have the recurrence for  $X_n$  as follows

$$X_{n+2} \stackrel{d}{=} I \left( A_n^\alpha X_{U_n}^{(1)} + B_n^\alpha X_{n+1-U_n}^{(2)} \right) + J \max \left( A_n^\alpha X_{U_n}^{(1)}, B_n^\alpha X_{n+1-U_n}^{(2)} \right), n \geq 1, \quad X_1 = 1, X_2 = 0, \quad (1.2)$$

where

$$A_n = \frac{U_n}{n+2} \text{ and } B_n = \frac{n+1-U_n}{n+2}.$$

Suppose that  $(X_n)$  converges, then limiting distribution  $Z$  of  $Y_n/n^\alpha$  satisfies the following fixed-point equation

$$Z \stackrel{d}{=} I \left( U^\alpha Z^{(1)} + (1-U)^\alpha Z^{(2)} \right) + J \max \left( U^\alpha Z^{(1)}, (1-U)^\alpha Z^{(2)} \right), \quad Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} Z. \quad (1.3)$$

We aim to find solutions  $(Z, \alpha)$  to 1.3 such that  $\mathcal{L}(Z) \neq \delta_0$ . Let us work with the space  $\mathcal{D}_2(1)$  of distributions over  $\mathbb{R}_+$  with mean 1 and finite second moment. Recall the  $p$ -Wasserstein metric  $d_p$  on  $\mathcal{D}_2(1)$  for  $p \in [1, 2]$

$$d_p(\mu, \nu) = \inf \left\{ \mathbb{E} [(X - Y)^p]^{1/p} : X \sim \mu, Y \sim \nu \right\}.$$

That means for every  $X \sim \mu$  and  $Y \sim \nu$ , we have  $d_p(\mu, \nu) \leq \mathbb{E} [(X - Y)^p]^{1/p}$ .

We give some properties of the fixed-point equation 1.3. Define the operator  $T_\alpha : \mathcal{D}_2(1) \rightarrow \mathcal{D}_2(1)$  according to the right-hand side of 1.3 as

$$T_\alpha(\mu) = \mathcal{L} \left( I \left( U^\alpha Z^{(1)} + (1-U)^\alpha Z^{(2)} \right) + J \max \left( U^\alpha Z^{(1)}, (1-U)^\alpha Z^{(2)} \right) \right), \quad Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} \mu. \quad (1.4)$$

**Lemma 1.** For every  $\mu, \nu \in \mathcal{D}_2(1)$  and  $\alpha \in (0, 1]$ , we have

$$d_2(T_\alpha(\mu), T_\alpha(\nu)) \leq \sqrt{\frac{2(u+v)}{2\alpha+1}} d_2(\mu, \nu) \text{ and } d_1(T_\alpha(\mu), T_\alpha(\nu)) \leq \frac{2(u+v)}{\alpha+1} d_1(\mu, \nu).$$

*Proof.* Since  $I$  and  $J$  are Bernoulli, we have  $I^2 = I$  and  $J^2 = J$ . Also, from  $I + J \leq 1$ , we have  $IJ = 0$ . Therefore, for every  $X, X^{(1)}, X^{(2)} \stackrel{\text{iid}}{\sim} \mu$  and  $Y, Y^{(1)}, Y^{(2)} \stackrel{\text{iid}}{\sim} \nu$ , we have

$$\begin{aligned} d_2^2(T_\alpha(\mu), T_\alpha(\nu)) &\leq \mathbb{E} \left[ I \left( \left( U^\alpha (X^{(1)} - Y^{(1)}) + (1-U)^\alpha (X^{(2)} - Y^{(2)}) \right) \right)^2 \right] \\ &\quad + \mathbb{E} \left[ J \left( \max \left( U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right) - \max \left( U^\alpha Y^{(1)}, (1-U)^\alpha Y^{(2)} \right) \right)^2 \right] \\ &:= A + B. \end{aligned}$$

For the first expectation  $A$ , we use  $\mathbb{E}[X] = \mathbb{E}[Y] = 1$  and get

$$\begin{aligned} A &= u \mathbb{E} \left[ U^{2\alpha} (X^{(1)} - Y^{(1)})^2 + (1-U)^{2\alpha} (X^{(2)} - Y^{(2)})^2 \right] \\ &= \frac{2u}{2\alpha+1} \mathbb{E}[(X - Y)^2]. \end{aligned}$$

For the second one,  $B$ , we use the inequality

$$(\max(a, b) - \max(c, d))^2 \leq \max(a - c, b - d)^2 \leq (a - c)^2 + (b - d)^2$$

for real numbers  $a, b, c, d$  and get

$$\begin{aligned} B &\leq v \mathbb{E} \left[ U^{2\alpha} (X^{(1)} - Y^{(1)})^2 + (1-U)^{2\alpha} (X^{(2)} - Y^{(2)})^2 \right] \\ &= \frac{2v}{2\alpha+1} \mathbb{E}[(X - Y)^2]. \end{aligned}$$

Therefore,

$$d_2^2(T_\alpha(\mu), T_\alpha(\nu)) \leq \frac{2(u+v)}{2\alpha+1} \mathbb{E}[(X - Y)^2].$$

Taking the expectation over  $X \sim \mu$  and  $Y \sim \nu$ , we get the desired result. The case for  $d_1$  is similar.  $\square$

For  $\mu \in \mathcal{D}_2(1)$  and  $\alpha \in (0, 1]$ , define

$$f(\mu, \alpha) = \mathbb{E}_{Y \sim T_\alpha(\mu)}[Y] = \frac{2u}{\alpha+1} + v \mathbb{E}_{Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} \mu} \left[ \max \left( U^\alpha Z^{(1)}, (1-U)^\alpha Z^{(2)} \right) \right] \quad (1.5)$$

**Lemma 2.** For every  $\mu \in \mathcal{D}_2(1)$ , there exists a unique  $\alpha \in (0, 1]$  such that  $f(\mu, \alpha) = 1$ .

*Proof.* It is clear that  $f$  is strictly decreasing in  $\alpha$ . Next, we show that  $f$  is continuous. The first term is continuous. We show that the second term is Lipschitz continuous. Let  $\alpha, \alpha' \in (0, 1]$

and assume without loss of generality that  $\alpha > \alpha'$ . We have

$$\begin{aligned}
& \mathbb{E} \left[ \max \left( U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right) - \max \left( U^{\alpha'} X^{(1)}, (1-U)^{\alpha'} X^{(2)} \right) \right] \\
& \leq \mathbb{E} \left[ \max \left( \left| U^\alpha - U^{\alpha'} \right| X^{(1)}, \left| (1-U)^\alpha - (1-U)^{\alpha'} \right| X^{(2)} \right) \right] \\
& \leq \mathbb{E} \left[ \left| U^\alpha - U^{\alpha'} \right| \right] + \mathbb{E} \left[ \left| (1-U)^\alpha - (1-U)^{\alpha'} \right| \right] \\
& = 2\mathbb{E} \left[ \left| U^\alpha - U^{\alpha'} \right| \right] \quad (U \stackrel{d}{=} 1-U) \\
& = 2\mathbb{E} \left[ U^{\alpha'} - U^\alpha \right] \quad (U^\alpha \geq U^{\alpha'}, a.s.) \\
& = \frac{2}{\alpha' + 1} - \frac{2}{\alpha + 1} = \frac{2(\alpha - \alpha')}{(\alpha + 1)(\alpha' + 1)} \\
& \leq 2(\alpha - \alpha').
\end{aligned}$$

Finally, we evaluate  $f$  at the endpoints.

$$\begin{aligned}
\lim_{\alpha \rightarrow 0+} f(\mu, \alpha) &= 2u + v \mathbb{E} \left[ \max \left( X^{(1)}, X^{(2)} \right) \right] > 2u + v > 1, \\
f(1) &= u + v \mathbb{E} \left[ \max \left( U^\alpha X^{(1)}, (1-U)^\alpha X^{(2)} \right) \right] \leq u + v \leq 1.
\end{aligned}$$

□

Consider  $\alpha : \mathcal{D}_2(1) \rightarrow (0, 1]$  as a function that maps a distribution  $\mu$  to the unique  $\alpha$  such that  $f(\mu, \alpha) = 1$ . By Lemma 2, this function is well-defined. That motivates an algorithm to find  $\alpha^*$  and the limiting distribution  $\mu^*$  of  $X_n$  as follows.

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**Algorithm 1**

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- 1: Initialize  $Z_1 = 1$  and  $\alpha_1$  solves  $\alpha + v2^{-\alpha} = 2(u + v) - 1$ .
  - 2: **for**  $k = 1, 2, \dots$  **do**
  - 3:   Compute  $Z_{n+1} = T_{\alpha_n}(\mathcal{L}(Z_n))$ .
  - 4:   Find  $\alpha_{n+1}$  such that  $\mathbb{E}[Z_{n+1}] = 1$  i.e.  $f(\mathcal{L}(Z_{n+1}), \alpha_{n+1}) = 1$ .
  - 5: **endfor**
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Using  $\max(a, b)^2 \leq a^2 + b^2$ , we bound the second moments as follows

$$\begin{aligned}
Z_{n+1}^2 &\leq I \left( U^{2\alpha_n} \left( Z_n^{(1)} \right)^2 + (1-U)^{2\alpha_n} \left( Z_n^{(2)} \right)^2 + 2(U(1-U))^{\alpha_n} Z_n^{(1)} Z_n^{(2)} \right) \\
&\quad + J \left( U^{2\alpha_n} \left( Z_n^{(1)} \right)^2 + (1-U)^{2\alpha_n} \left( Z_n^{(2)} \right)^2 \right) \\
&= (I + J) \left( U^{2\alpha_n} \left( Z_n^{(1)} \right)^2 + (1-U)^{2\alpha_n} \left( Z_n^{(2)} \right)^2 \right) + 2I(U(1-U))^{\alpha_n} Z_n^{(1)} Z_n^{(2)}
\end{aligned}$$

Taking the expectation, we get

$$\mathbb{E} [Z_{n+1}^2] \leq \frac{2(u+v)}{2\alpha_n + 1} \mathbb{E} [Z_n^2] + 2uB(\alpha_n + 1, \alpha_n + 1),$$

where  $B$  is the beta function. The function  $\alpha \mapsto B(\alpha + 1, \alpha + 1)$  is bounded on  $(0, 1]$ . In particular, we have  $B(\alpha + 1, \alpha + 1) \in [1/6, 1)$ . For  $\sup_n \mathbb{E}[Z_n^2] < \infty$ , at least we need some  $k$  such that

$$\frac{2(u+v)}{2\alpha_k + 1} < 1 \iff \alpha_k > u + v - \frac{1}{2}$$

We may choose to impose  $\alpha_1 > u + v - \frac{1}{2}$  and use  $\alpha_1 > 2u + v - 1$  to derive the condition  $u \geq \frac{1}{2}$ .

## 1.2 Existence of a Fixed Point

Let us define the operator  $\Phi : \mathcal{D}_2(1) \rightarrow \mathcal{D}_2(1)$  as

$$\Phi(\mu) = T_{\alpha(\mu)}(\mu). \quad (1.6)$$

Let  $\mathcal{D}_2(1, M)$  be the subset of  $\mathcal{D}_2(1)$  with second moment bounded by  $M$

$$\mathcal{D}_2(1, M) = \left\{ \mu \in \mathcal{D}_2(1) : \int_0^\infty x^2 d\mu \leq M \right\}.$$

**Lemma 3.** *There exists a real number  $M$  such that the operator  $\Phi$  defined in 1.6 maps  $\mathcal{D}_2(1, M)$  into itself.*

*Proof.* Let  $\mu \in \mathcal{D}_2(1)$  have finite second moment. Let  $Z \sim \mu$ . Follow the steps in the previous section, we have

$$\begin{aligned} \mathbb{E} [\Phi(Z)^2] &\leq \frac{2(u+v)}{2\alpha+1} \mathbb{E} [Z^2] + 2uB(\alpha+1, \alpha+1) \\ &< \frac{2(u+v)}{2(2u+v)-1} \mathbb{E} [Z^2] + 2uB(2u+v, 2u+v) \end{aligned}$$

Choosing

$$M = B(2u+v, 2u+v) \frac{2u(2u-1)}{2(2u+v)-1} \text{ and } \mathbb{E}[Z^2] \leq M,$$

we have  $\mathbb{E} [\Phi(Z)^2] \leq M$ . Hence,  $\Phi$  maps  $\mathcal{D}_2(1, M)$  into itself.  $\square$

We attempt to use the Tychonoff fixed-point theorem (See the Discussion section for why Banach or Schauder fixed-point theorems do not apply here).

**Theorem 1** (Tychonoff [4, Cor. 9.6]). *Let  $\mathcal{C}$  be a nonempty, compact, convex subset of a locally convex space  $\mathcal{M}$ . Let  $f : \mathcal{C} \rightarrow \mathcal{C}$  be a continuous mapping. Then  $f$  has a fixed point.*

The host space  $\mathcal{M}$  is the space of signed Borel measures on  $\mathbb{R}$ , equipped with the topology of weak convergence. Details of  $\mathcal{M}$  being locally convex are provided in the Appendix.

Then  $\Phi : \mathcal{D}_2(1, M) \rightarrow \mathcal{D}_2(1, M)$ .

**Lemma 4.** *The subset  $\mathcal{D}_2(1, M)$  of  $\mathcal{M}$  is nonempty, compact, and convex.*

*Proof.* The set  $\mathcal{D}_2(1, M)$  is nonempty since it contains  $\delta_1$ . For convexity, let  $\mu, \nu \in \mathcal{D}_2(1, M)$ . Then for every  $\lambda \in [0, 1]$ , we have

$$\begin{aligned} \int_0^\infty x d(\lambda\mu + (1-\lambda)\nu) &= \lambda \int_0^\infty x d\mu + (1-\lambda) \int_0^\infty x d\nu = 1, \\ \int_0^\infty x^2 d(\lambda\mu + (1-\lambda)\nu) &= \lambda \int_0^\infty x^2 d\mu + (1-\lambda) \int_0^\infty x^2 d\nu \leq M. \end{aligned}$$

For compactness, we need to show that  $\mathcal{D}_2(1, M)$  is tight (cf. Definition ??) and closed. For tightness, we use Markov's inequality. For any  $\mu \in \mathcal{D}_2(1, M)$  and any  $R > 0$

$$\mu(\{x \geq R\}) \leq \frac{1}{R^2} \int_0^\infty x^2 d\mu \leq \frac{M}{R^2}. \quad (1.7)$$

For closedness, let  $(\mu_n)$  be a sequence in  $\mathcal{D}_2(1, M)$  that converges weakly to  $\mu$ . We show that  $\mu \in \mathcal{D}_2(1, M)$ . The boundedness of the second moment follows from a generalized Fatou's lemma [2, Theorem 1.1]

$$\int_0^\infty x^2 d\mu \leq \liminf_{n \rightarrow \infty} \int_0^\infty x^2 d\mu_n \leq M.$$

The function  $x \mapsto x$  is uniformly integrable with respect to  $\mathcal{D}_2(1, M)$  as follows. For any  $\mu \in \mathcal{D}_2(1, M)$  and any  $R > 0$

$$\int_R^\infty x d\mu \leq \int_R^\infty x \left(\frac{x}{R}\right) d\mu = \frac{1}{R} \int_R^\infty x^2 d\mu \leq \frac{1}{R} \int_0^\infty x^2 d\mu \leq \frac{M}{R}.$$

which tends to 0 as  $R \rightarrow \infty$ . Therefore, the first moment converges [1, Theorem 3.5]

$$\int_0^\infty x d\mu_n \rightarrow \int_0^\infty x d\mu.$$

□

**Remark.** Since  $x \mapsto x$  is uniformly integrable with respect to  $\mathcal{D}_2(1, M)$ , weak convergence in  $\mathcal{D}_2(1, M)$  is equivalent to convergence in  $d_1$  [3, Theorem 6.9].

**Lemma 5.** *The operator  $\Phi$  defined in 1.6 is continuous on  $\mathcal{D}_2(1, M)$  with respect to the weak topology.*

*Proof.* Let  $(\mu_n)$  be a sequence in  $\mathcal{D}_2(1, M)$  that converges weakly to  $\mu$ , which is equivalent to that  $d_1(\mu_n, \mu) \rightarrow 0$ . To show that  $\Phi(\mu_n) \xrightarrow{w} \Phi(\mu)$ , we will show that  $d_1(\Phi(\mu_n), \Phi(\mu)) \rightarrow 0$  by showing that  $\Phi$  is Lipschitz with respect to  $d_1$ . Let  $\mu, \nu \in \mathcal{D}_2(1, M)$ . Shorten  $\alpha_1 := \alpha(\mu)$  and  $\alpha_2 := \alpha(\nu)$ . Without loss of generality, assume that  $\alpha_1 > \alpha_2$ . We have

$$\begin{aligned} d_1(\Phi(\mu), \Phi(\nu)) &= d_1(T_{\alpha_1}(\mu), T_{\alpha_2}(\nu)) \\ &\leq d_1(T_{\alpha_2}(\mu), T_{\alpha_2}(\nu)) + d_1(T_{\alpha_1}(\mu), T_{\alpha_2}(\mu)). \end{aligned}$$

By Lemma 1, we have

$$d_1(T_{\alpha_2}(\mu), T_{\alpha_2}(\nu)) \leq \frac{2(u+v)}{\alpha_2+1} d_1(\mu, \nu) \leq 2(u+v) d_1(\mu, \nu).$$

For the second distance, we let  $Y^{(1)}, Y^{(2)} \sim \nu$ . Then, following the same steps as in Lemma 2, we have

$$d_1(T_{\alpha_1}(\mu), T_{\alpha_2}(\mu)) \leq 2(u+v)(\alpha_1 - \alpha_2).$$

We have  $f(\mu, \alpha_1) = f(\nu, \alpha_2) = 1$ . Hence

$$|f(\mu, \alpha_1) - f(\mu, \alpha_2)| = |f(\mu, \alpha_2) - f(\nu, \alpha_2)|.$$

Consider the left-hand side term where we fix the distribution  $\mu$ . We rewrite  $f$  as

$$f(\mu, \alpha) = \frac{2u}{\alpha+1} + v\mathbb{E} \left[ U^\alpha Z^{(1)} \mathbf{1}_A + (1-U)^\alpha Z^{(2)} \mathbf{1}_{A^c} \right], \quad Z^{(1)}, Z^{(2)} \stackrel{\text{iid}}{\sim} \mu,$$

where  $A$  is the event that  $U^\alpha Z^{(1)} \geq (1 - U)^\alpha Z^{(2)}$ . Since  $\mathbb{P}(U^\alpha Z^{(1)} = (1 - U)^\alpha Z^{(2)}) = 0$ , we have

$$\frac{\partial}{\partial \alpha} f(\mu, \alpha) = -\frac{2u}{(\alpha + 1)^2} + v\mathbb{E} \left[ \ln(U)U^\alpha Z^{(1)} \mathbf{1}_A + \ln(1 - U)(1 - U)^\alpha Z^{(2)} \mathbf{1}_{A^c} \right].$$

Since  $\ln(U) < 0$  and  $\ln(1 - U) < 0$ , we have for every  $\alpha \in (0, 1]$ ,

$$\left| \frac{\partial}{\partial \alpha} f(\mu, \alpha) \right| \geq \frac{2u}{(\alpha + 1)^2}.$$

By Mean Value Theorem, we have

$$|f(\mu, \alpha_1) - f(\mu, \alpha_2)| \geq \frac{2u}{(\alpha_1 + 1)^2} |\alpha_1 - \alpha_2| \geq \frac{u}{2} |\alpha_1 - \alpha_2|$$

We also have

$$|f(\mu, \alpha_2) - f(\nu, \alpha_2)| \leq \frac{2v}{\alpha_2 + 1} d_1(\mu, \nu) \leq 2vd_1(\mu, \nu).$$

Thus,

$$d_1(\Phi(\mu), \Phi(\nu)) \leq 2(u + v) \left( 1 + \frac{4v}{u} \right) d_1(\mu, \nu).$$

□

**Theorem 2.** *The operator  $\Phi$  defined in 1.6 has a fixed point in  $\mathcal{D}_2(1, M)$ . That is, there exists a pair  $\alpha^* \in (0, 1]$  and  $\mu^* \in \mathcal{D}_2(1, M)$  satisfying (1.3) with  $Z \sim \mu^*$ .*

*Proof.* The theorem follows from Tychonoff's fixed-point theorem, Lemma 4, and Lemma 5. □

### 1.3 Uniqueness of the Fixed Point

For uniqueness, the intuition is as follows. Algorithm 1 only converges if we adjust  $\alpha_n$  at each step to ensure that the mean of  $\int x d\mu_n = 1$ . We started from the most concentrated distribution  $\mu_1 = \delta_1$  and increased  $\alpha_n$  to fight against the variance of  $\delta_n$ . Hence, if we fix  $\alpha^*$ , then the sequence  $(\mu_n)$  will converge to  $\delta_0$ . Otherwise, if we fix  $\alpha = \alpha^* - \epsilon$ , then the sequence will diverge to  $\infty$ .

If a single  $\alpha^*$  gives two distributions  $\mu_1^*$  and  $\mu_2^*$  that are fixed points of  $T_{\alpha^*}$ , then from Lemma 1, we have

$$d_2(\mu_1^*, \mu_2^*) \leq \sqrt{\frac{2(u + v)}{2\alpha + 1}} d_2(\mu_1^*, \mu_2^*).$$

Hence  $\mu_1^* = \mu_2^*$ .

**Lemma 6.** *Let  $(\alpha^*, \mu^*)$  be a solution to (1.3). Then  $T_{\alpha^*}^n(\delta_1) \xrightarrow{d} \delta_0$ .*

### 1.4 Convergence of the Algorithm

# References

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